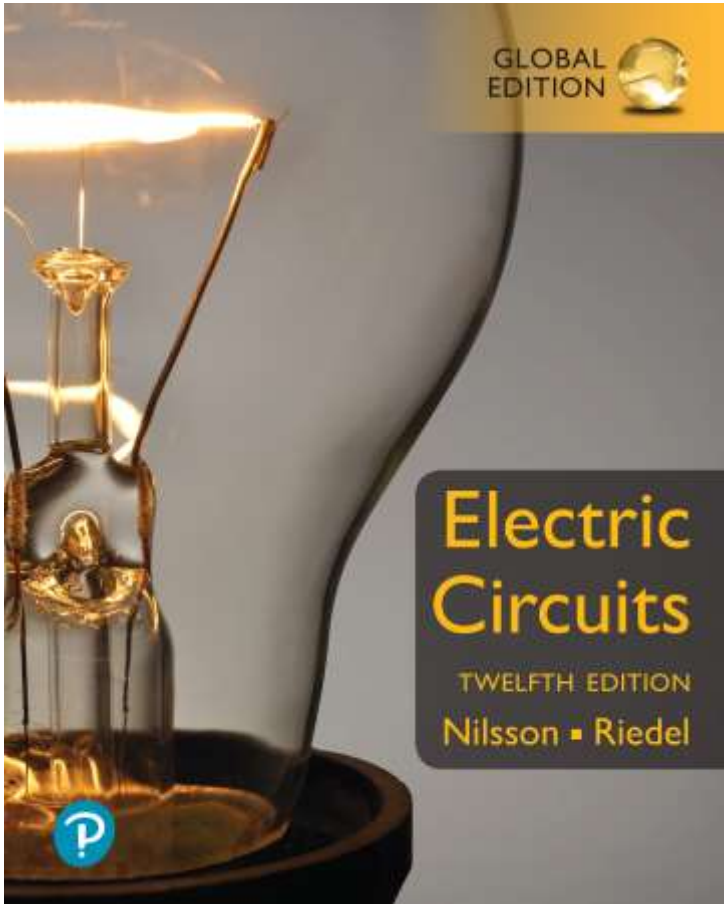
Electric Circuits

Twelfth Edition, Global Edition



Chapter 4 – Part 8

Techniques of Circuit Analysis

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Chapter 4 – techniques of circuit analysis (1 of 16)

Condition for maximum power transfer

Suppose we can vary the load resistance. For what value of load resistance will maximum power be absorbed by the load?

$$p_L = i^2 R_L; \qquad i = \frac{V_{Th}}{R_{Th} + R_L}$$

$$\therefore \qquad p_L = \frac{R_L V_{Th}^2}{(R_{Th} + R_L)^2}$$

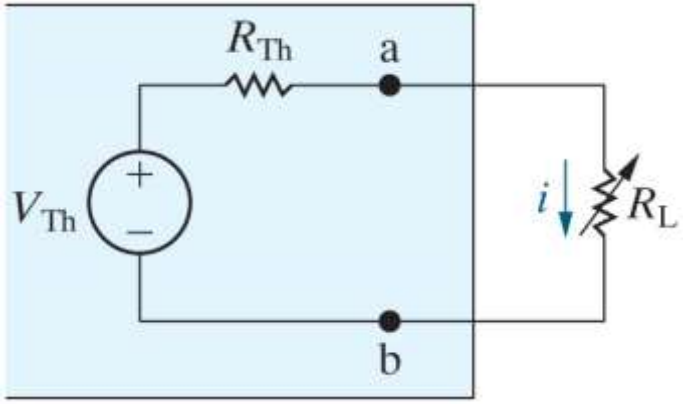
For max. power, $\frac{dp_L}{dR_L} = 0$

$$\frac{dp_L}{dR_L} = \frac{V_{Th}^2}{(R_{Th} + R_L)^2} - \frac{2R_L V_{Th}^2}{(R_{Th} + R_L)^2}$$

$$= \frac{V_{Th}^2 (R_{Th} + R_L) - 2R_L V_{Th}^2}{(R_{Th} + R_L)^3} = 0$$

$$\Rightarrow \qquad V_{Th}^2 (R_{Th} + R_L) - 2R_L V_{Th}^2 = 0$$

$$\Rightarrow \qquad (R_{Th} + R_L) = 2R_L \qquad \therefore \qquad R_L = R_{Th}$$



Represents the Thevenin equivalent of an arbitrary subcircuit that will not change

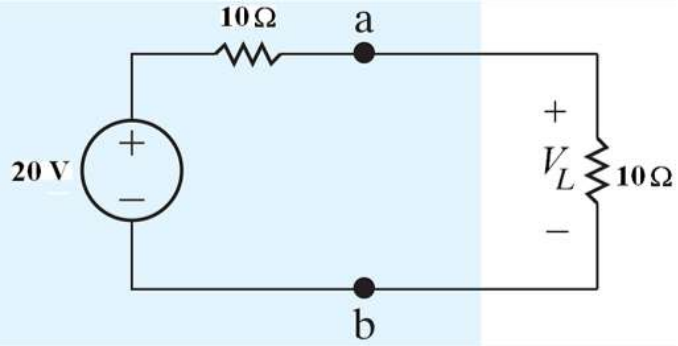
Represents a variable load

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Chapter 4 – techniques of circuit analysis (2 of 16)

In the circuit below, the load resistor is matched to the Thevenin resistance – how much voltage drops across the load resistor?



A. 20 V

B. 10 V

C. 5 V

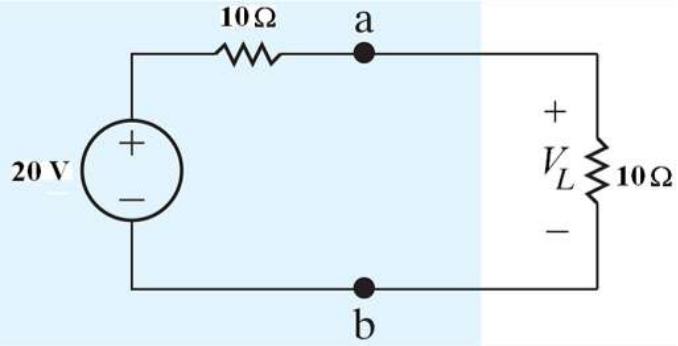
D. 40 V

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Chapter 4 – techniques of circuit analysis (2 of 16)

In the circuit below, the load resistor is matched to the Thevenin resistance – how much voltage drops across the load resistor?



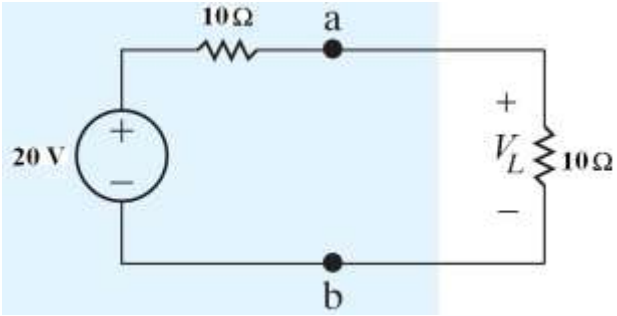
- ✗ A. 20 V
- ✓ B. 10 V
- ✗ C. 5 V
- ✗ D. 40 V

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Chapter 4 – techniques of circuit analysis (3 of 16)

In the circuit below, the load resistor is matched to the Thevenin resistance – how much power is absorbed by the load resistor?



A. 100 W

B. 50 W

C. 20 W

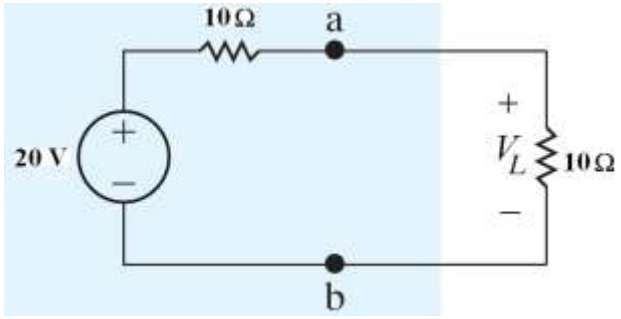
D. 10 W

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Chapter 4 – techniques of circuit analysis (3 of 16)

In the circuit below, the load resistor is matched to the Thevenin resistance – how much power is absorbed by the load resistor?



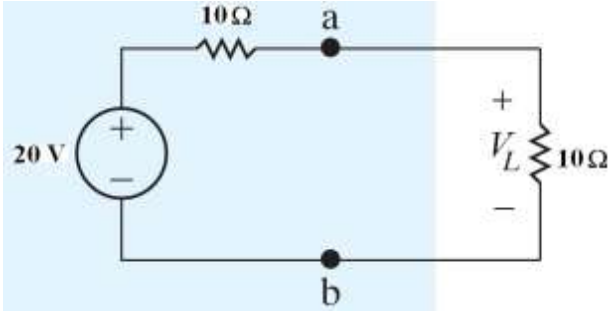
- ✗ A. 100 W
- ✗ B. 50 W
- ✗ C. 20 W
- ✓ D. 10 W

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Chapter 4 – techniques of circuit analysis (4 of 16)

In the circuit below, the load resistor is matched to the Thevenin resistance – how much power is delivered by the voltage source?



A. 100 W

B. 50 W

C. 20 W

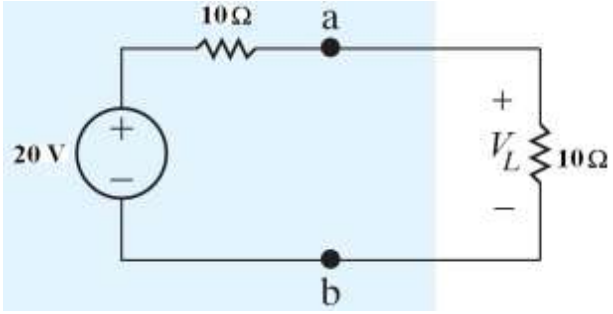
D. 10 W

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Chapter 4 – techniques of circuit analysis (4 of 16)

In the circuit below, the load resistor is matched to the Thevenin resistance – how much power is delivered by the voltage source?



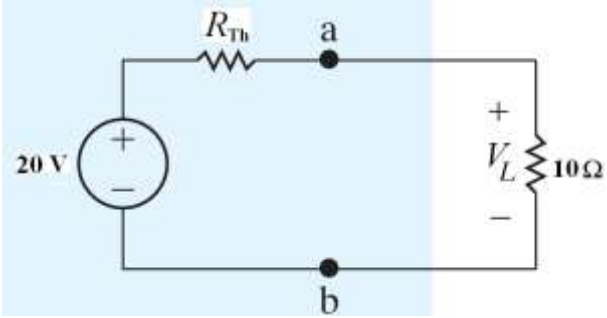
- ✗ A. 100 W
- ✗ B. 50 W
- ✓ C. 20 W
- ✗ D. 10 W

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Chapter 4 – techniques of circuit analysis (5 of 16)

Suppose that we vary the Thevenin resistor instead of the load resistor. For what value of Thevenin resistance is maximum power delivered to the load resistance?



A. 20Ω

B. 10Ω

C. 0Ω

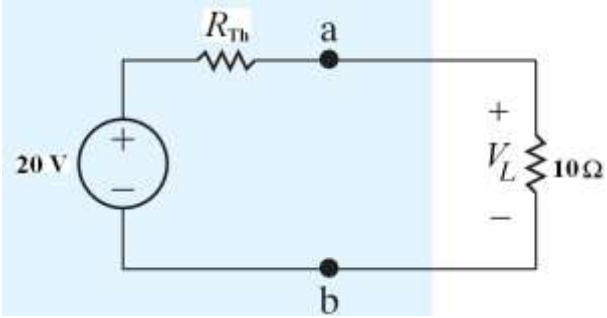
D. None of the above

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Chapter 4 – techniques of circuit analysis (5 of 16)

Suppose that we vary the Thevenin resistor instead of the load resistor. For what value of Thevenin resistance is maximum power delivered to the load resistance?



- ✗ A. 20Ω
- ✗ B. 10Ω
- ✓ C. 0Ω
- ✗ D. None of the above

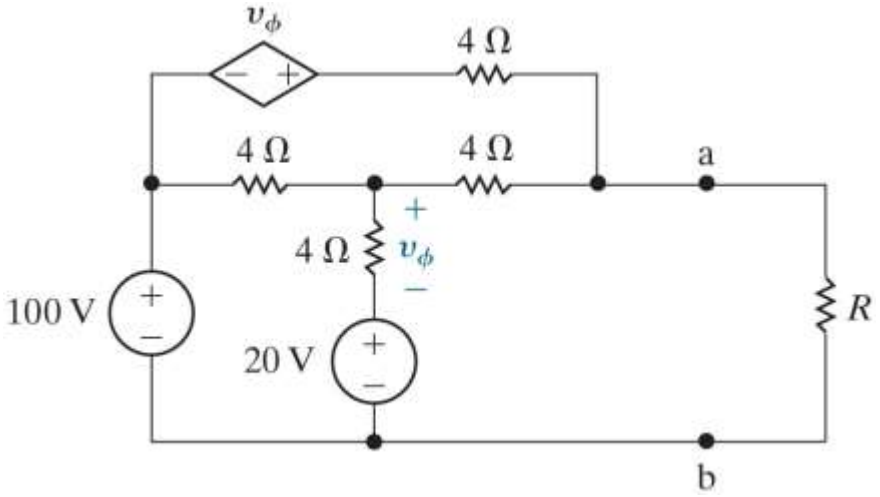
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Chapter 4 – techniques of circuit analysis (6 of 16)

Example of maximum power transfer

Find the value of *R* such that maximum power is transferred to *R*, and find that maximum power.



Approach – find the open circuit and the short circuit current by analyzing two circuits – see the next slides.

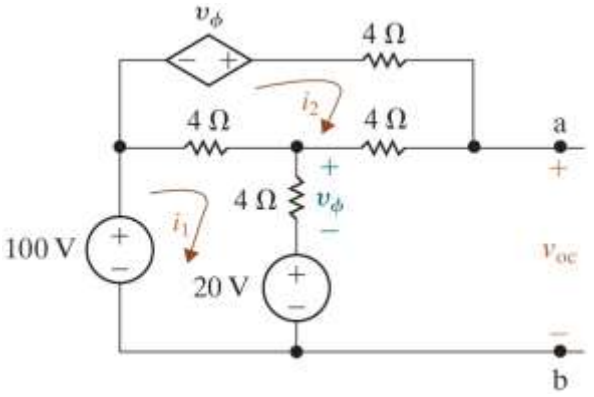
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Chapter 4 – techniques of circuit analysis (7 of 16)

Find the value of *R* such that maximum power is transferred to *R*, and find that maximum power.

Find the open circuit voltage:



i_1 mesh: $-100 + 4(i_1 - i_2) + 4i_1 + 20 = 0$

i_2 mesh: $-v_\varphi + 4i_2 + 4(i_2 - i_1) = 0$

One $4i_2$ is missing in the equation. (Thanks to Alper for pointing it out.)

constraint: $v_\varphi = 4i_1$

Solving: $i_1 = 15$ A; $i_2 = 10$ A; $v_\varphi = 60$ V

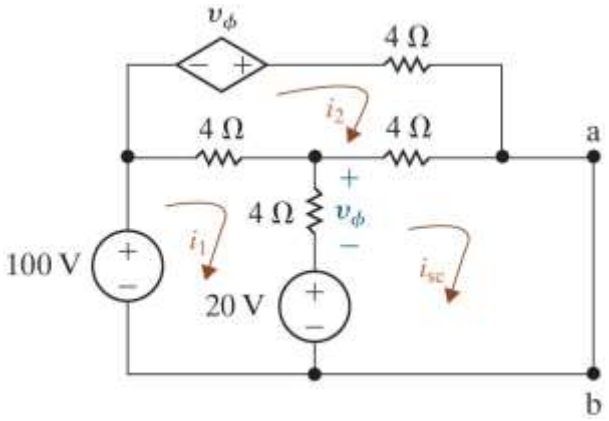
$\therefore \quad v_{oc} = 4i_2 + 4i_1 + 20 = 120$ V

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Chapter 4 – techniques of circuit analysis (8 of 16)

Find the value of *R* such that maximum power is transferred to *R*, and find that maximum power.



Find the short circuit current:

i_1 mesh: $-100 + 4(i_1 - i_2) + 4(i_1 - i_{sc}) + 20 = 0$

i_2 mesh: $-v_\varphi + 4i_2 + 4(i_2 - i_{sc}) + 4(i_2 - i_1) = 0$

i_{sc} mesh: $-20 + 4(i_{sc} + i_1) + 4(i_{sc} - i_2) = 0$

constraint: $v_\varphi = 4(i_1 - i_{sc})$

Solving: $i_1 = 45$ A; $i_2 = 30$ A; $i_{sc} = 40$ A; $v_\varphi = 20$ V

$$\therefore \quad R_{Th} = \frac{v_{oc}}{i_{sc}} = \frac{120}{40} = 3\Omega$$

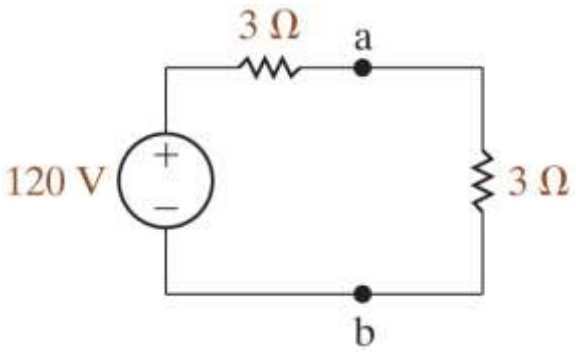
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Chapter 4 – techniques of circuit analysis (9 of 16)

Find the value of *R* such that maximum power is transferred to *R*, and find that maximum power.

The Thevenin equivalent:



$$p_L = \frac{60^2}{3} = 1200 \text{ W}$$

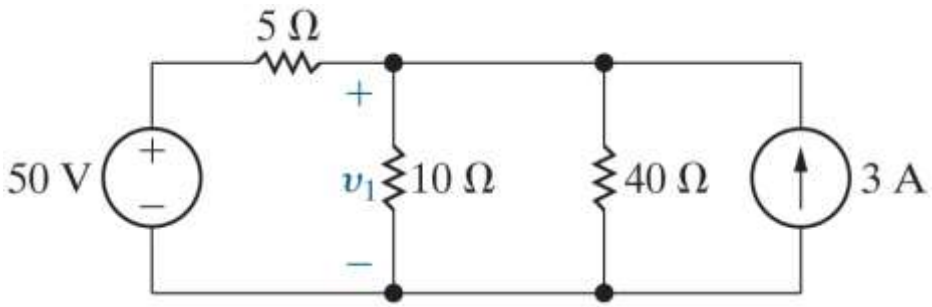
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Chapter 4 – techniques of circuit analysis (10 of 16)

Linear circuits, which are examples of linear systems, obey the property of superposition. Thus, when a linear circuit has more than one independent source, we can find that circuit's response by summing the responses to each source independently.

Use superposition to find the voltage v_1 in the circuit here.

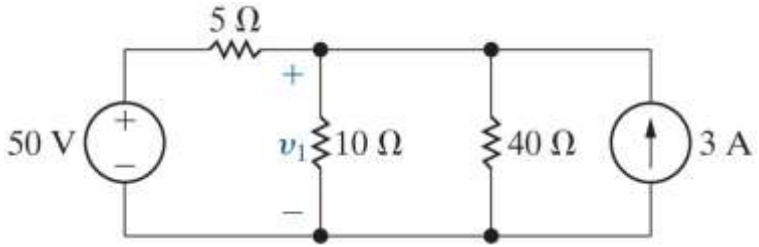


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Chapter 4 – techniques of circuit analysis (11 of 16)

We begin by calculating the voltage across the 10Ω resistor due to the voltage source. We need to set the value of the current source to 0 A. Therefore, we replace the current source with a(n)

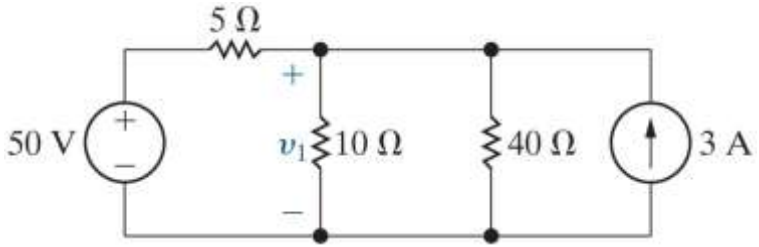


A. open circuit
B. short circuit
C. voltage source
D. dependent source

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Chapter 4 – techniques of circuit analysis (11 of 16)

We begin by calculating the voltage across the 10Ω resistor due to the voltage source. We need to set the value of the current source to 0 A. Therefore, we replace the current source with a(n)



- ✔ A. open circuit
- ✘ B. short circuit
- ✘ C. voltage source
- ✘ D. dependent source

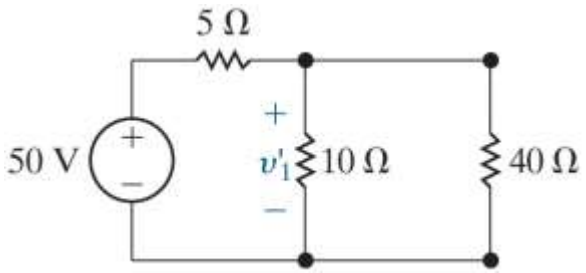
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Chapter 4 – techniques of circuit analysis (12 of 16)

Use superposition to find the voltage v_1 in the circuit here.

Calculate the response v_1' to the voltage source:



Combine the 10Ω and 40Ω resistors into a single equivalent resistor. Then use voltage division to find v_1'.

$$10 \,||\, 40 = 8\ \Omega;$$

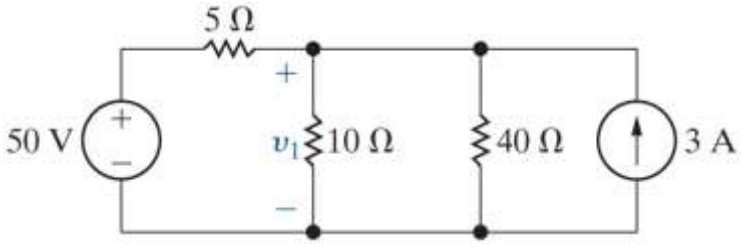
$$v_1' = \frac{8}{8+5}(50) = 30.77\ \text{V}$$

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Chapter 4 – techniques of circuit analysis (13 of 16)

Now we calculate the voltage across the 10Ω resistor due to the current source. We need to set the value of the voltage source to 0 V. Therefore, we replace the voltage source with a(n)



- ✗ A. open circuit
- ✓ B. short circuit
- ✗ C. voltage source
- ✗ D. dependent source

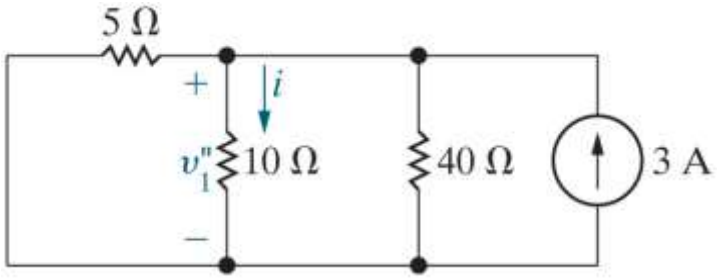
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Chapter 4 – techniques of circuit analysis (14 of 16)

Use superposition to find the voltage v_1 in the circuit here.

Calculate the response v_1'' to the current source:



Use current division to find the current i in the 10Ω resistor.

Then use Ohm's law to find v_1''.

$$i = \frac{5 \,||\, 10 \,||\, 40}{10}(3) = \frac{3.077}{10}(3) = 0.923 \text{ A};$$

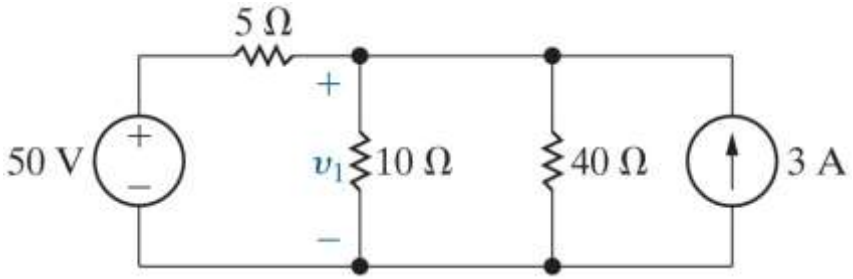
$$v_1'' = 10i = 9.23 \text{ V}$$

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Chapter 4 – techniques of circuit analysis (15 of 16)

Use superposition to find the voltage v_1 in the circuit here.



Finally, sum the responses to each source independently to get the total response.

$$v_1 = v_1' + v_1'' = 30.77 + 9.23 = 40 \text{ V}$$

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Chapter 4 – techniques of circuit analysis (16 of 16)

Summary of circuit analysis techniques:

- Ohm’s Law
- KVL and KCL
- Combining resistors in series and parallel
- Voltage division and current division
- Node voltage method
- Mesh current method
- Source transformation
- Thevenin and Norton equivalents
- Superposition

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